went on to explain how if groups within a species become isolated from the rest of the population, they may start to differ from the rest, and over time, through genetic drift and natural selection, may even evolve into new species.

Modern technological advances, including electron microscopy and mitochondrial DNA analysis, have revealed much information—some of it surprising—about the number of species their relationships. In 1966, striving to reflect the intricacies of evolution, German entomologist Willi Hennig proposed a new taxonomic system of clades—groups of organisms based on a common ancestor. In the 1970s, US biologist Carl Woese classified all life into three new domains. As of 2018, about 1.74 million extant plant and animal species have been described, but estimates of the total number range from 2 million to 1 trillion.

The threat to diversity

By the late 20th century, however, alongside a growing knowledge of the scale and critical role of biodiversity—and of how evolution can destroy species as well as create them—American ecologist E.O. Wilson and others made the world aware that human activity was responsible for causing a rapid acceleration in the extinction rate. In 1964, to highlight these dangers, the International Union for Conservation of Nature (IUCN) established a Red List of Threatened Species, followed in 2014 by a similar list for ecosystems. Some have even warned that Earth could be on the verge of a sixth mass extinction. Many policies are now being proposed to counter this, including the protection of biodiversity hotspots.

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